

The Daily Engineer

"All the Commits Fit to Ship" — Reliability, Latency & the Quiet Pager

BENGALURU EDITION • APR 2026

CANDIDATE: PRAKHAR SHEKHAR PARTHASARTHI

FULL-STACK SWE

★ BREAKING • CAREER DESK ★

Ex-Amazon Engineer Ships Four Microservices, Cuts Latency 47%

Three years on the Travel desk: idempotent event-driven systems, Java-CDK infrastructure, and an on-call queue brought from 20+ tickets a shift to roughly five.

BY THE NEWSROOM • JAVA • SPRING BOOT • AWS • REACT • DISTRIBUTED SYSTEMS

47% P90 LATENCY CUT	+33% CONVERSION LIFT	195k REDIS GET RPS	20→5 ON-CALL / SHIFT
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The Lead Story

PROFILE • FROM THE CAREER DESK

A full-stack software engineer with 3+ years at Amazon across Payments and Travel, the candidate owned 4 customer-facing microservices end-to-end — design, build, deploy, and on-call.

The backend is Java-heavy: **Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture` and bounded `ExecutorService`, idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure authored entirely in **AWS CDK (Java)** — paired with **React (SSR & CSR) + Redux** on customer-facing flows.

CAREER • EXPERIENCE

Software Development Engineer

AMAZON • TRAVEL (HOTELS & FLIGHTS) • JUL 2022 – SEP 2025 • BENGALURU

- Owned 4 microservices (search, listing, room, review) on Spring Boot/MVC — REST + @Repository DynamoDB DAOs via Spring DI.
- JVM concurrency: `CompletableFuture.allOf()` on a bounded pool; 6 partner APIs parallelised — p90 1.1s→0.59s (-47%), conversion +33%.
- Event-driven: idempotent flight-schedule webhook — SQS + DynamoDB dedup + audit; Step Functions state machine, -43 bps CPT.
- IaC in AWS CDK (Java); migrated 2 services off CloudFormation — halved rollbacks.
- On-call mined to 20+→5/shift; 150+ incidents; 80%+ coverage.

SDE, Intern

AMAZON • 2021 • 2022 • BENGALURU

- Migrated GraphQL resolvers → Java (API Gateway + Lambda); shipped One-Click Renewal (async Java); Guice DI, 90%+ coverage.

CLASSIFIEDS • PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

RESP2 Redis built from scratch — bounded `ExecutorService`, lock-free reads, hybrid TTL eviction, TCP integration tests.

SET 154k • GET 195k • INCR 128k rps • p50 < 0.3ms • 42 tests

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Real-time markdown viewer — preview, syntax highlight, Mermaid, 10 themes, PDF/HTML export. Vanilla JS, sub-100KB.

155+ Vitest tests • Mermaid • export

RECORDS • EDUCATION

M.C.A., Computer Science

MNNIT Prayagraj GPA 9.25 • 2019–2022

B.Sc. (Hons), CS

University of Delhi • 2015–2018

Honors: Amazon Pay Merchants Categories Champion (2023) — 100+ trust-buster bugs.

The Technical Review

Full Stack, Tooling & the Engineering Method — Continued

★ SECTION B • THE TOOLBENCH ★

The Complete Stack, On the Record

JAVA & JVM

Java 17, Spring Boot, Spring MVC, Spring DI, Google Guice, JUnit 5, Mockito, SLF4J, Lombok, Maven, Gradle. Concurrency: `CompletableFuture`, `ExecutorService`, `ConcurrentHashMap`, thread-pool tuning, JVM profiling, GC analysis.

DISTRIBUTED SYSTEMS

Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry with exponential backoff, DLQ handling, at-least-once semantics, TDD.

AWS

Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java) — typed IaC for all service infrastructure.

FRONT-END

React.js (SSR & CSR), Redux (1.5 yrs production) on customer-facing Amazon flows; HTML / CSS, JavaScript / TypeScript.

DATA & STORAGE

DynamoDB, Redis, NoSQL modelling, SQL, caching strategies.

OTHER LANGUAGES

Python, C++, VTL.

TOOLING & PROCESS

Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile / Scrum, on-call, code review, incident response.

EDITORIAL • THE METHOD

"The shift you don't notice is the best one." The candidate's record favors idempotency over "just retry it," writes the runbook before the Slack update, and treats the on-call corpus as a backlog to automate — reclaiming roughly 2 SDE-weeks per week.

Additional dispatches on the record: GZIP on 150-200 MB catalog payloads cut wire size 95%, eliminating network-time-out failures; an auto-insurance purchase flow lifted renewals 60%+.

OP-ED • AVAILABILITY

Open to Senior SWE roles. Java / distributed systems / AWS. Reliability-first, ownership-minded, ships quiet.

"Refactored sequential downstream calls into `CompletableFuture.allOf()` — hotel landing p90 fell from 1.1s to 0.59s."

TIMELINE • ON THE RECORD

- 2025

Closed out Travel tenure — 4 services, 150+ incidents resolved.
- 2024

Idempotent flight-webhook shipped; -43 bps CPT.
- 2023

Concurrency refactor: -47% p90, +33% conversion · Merchants Champion.
- 2022

Promoted to SDE; owned hotel booking microservices.
- 2021

SDE Intern — GraphQL→Java migration.
- 2019

Began M.C.A. at MNNIT Prayagraj (GPA 9.25).